ON A NOETHERIAN REGULAR SCHEME THE K-THEORY OF COHERENT SHEAVES SUPPORTED ON A CLOSED SUBSCHEME AGREES WITH THE K-THEORY OF PERFECT COMPLEXES SUPPORTED THERE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FER (sources/stacks-chow/pinned-original-source.tex line 13003).

Allowed standard prerequisites (editorial).

- On a regular scheme perfect complexes coincide with bounded complexes with coherent cohomology. Derived Categories of Schemes, Lemma labelled perfect-lemma-perfect-on-regular (Stacks tag 0FDC), invoked at proof lines 13020-13022. Supplies D_perf(O_X) = D^b_Coh(O_X), so that the cohomology sheaves of an object of D_{Z,perf}(O_X) are coherent and only finitely many are nonzero.
- The two maps whose composite is studied, and the finite filtration of a coherent module by O_Z-modules. Same chapter, Remarks labelled remark-perf-Z-cohomology-K (source line 12945) and remark-perf-Z-regular (source line 12978), invoked in the statement. Supply the two maps in question; the truncation argument and the filtration step are carried out in the counted span.

Source reference. Original remark 0FEP: sources/stacks-chow/pinned-original-source.tex, source lines 12945–12945. Same-chapter pointer into the author environment remark at source line 12945; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original remark 0FEQ: sources/stacks-chow/pinned-original-source.tex, source lines 12978–12978. Same-chapter pointer into the author environment remark at source line 12978; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0FDC: sources/stacks-chow/tags.txt, source lines 19212–19212. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 13022 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let X be a Noetherian regular scheme. Let $Z \subset X$ be a closed subscheme. The maps constructed in Remarks 0FEP and 0FEQ are mutually inverse and we get $K'_0(Z) = K_0(D_{Z,perf}(\mathcal{O}_X))$.*

Proof. Clearly the composition

$$K_0(Coh(Z)) \longrightarrow K_0(D_{Z,perf}(\mathcal{O}_X)) \longrightarrow K_0(Coh(Z))$$

is the identity map. Thus it suffices to show the first arrow is surjective. Let E be an object of $D_{Z,perf}(\mathcal{O}_X)$. Recall that $D_{perf}(\mathcal{O}_X) = D^b_{Coh}(\mathcal{O}_X)$ by Derived Categories of Schemes, Lemma 0FDC. Hence the cohomologies $H^i(E)$ are coherent, can be viewed as objects of $D_{Z,perf}(\mathcal{O}_X)$, and only a finite number are nonzero. Using the distinguished triangles of canonical truncations the reader sees that

$$[E] = \sum (-1)^i [H^i(E)[0]]$$

in $K_0(D_{Z,perf}(\mathcal{O}_X))$. Then it suffices to show that $[\mathcal{F}[0]]$ is in the image of the map for any coherent $\mathcal{O}_X$-module set theoretically supported on Z. Since we can find a finite filtration on $\mathcal{F}$ whose subquotients are $\mathcal{O}_Z$-modules, the proof is complete. $\square$